

Topic 2.12 - Logarithmic Function Manipulation

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

Logarithmic properties transform multiplication into addition, division into subtraction, and powers into coefficients. These identities are valid only when every logarithm involved has a positive argument. A correct rewrite includes the same domain as the original expression.

Learning targets

- Expand and condense logarithmic expressions using product, quotient, and power properties.
- Use change of base and natural logarithms to evaluate and compare logarithms.
- Explain algebraic and graphical equivalences while preserving domain restrictions.

Language and structure

Term	Meaning in this topic
product property	$\log_b(MN) = \log_b M + \log_b N$ for $M, N > 0$.
quotient property	$\log_b(M/N) = \log_b M - \log_b N$ for $M, N > 0$.
power property	$\log_b(M^p) = p \log_b M$ when the expressions are defined.
condense	Combine several logarithms into one logarithm.
domain equivalence	A rewrite must preserve the set of allowed inputs, not only the algebraic appearance.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A student claims $\log(x + 4) = \log x + \log 4$. Test the claim at one convenient input and then state the correct property that may have caused the confusion.

Concept 1: Expand with structure

Core idea

Factor products, quotients, and powers before applying logarithmic properties.

- Keep numerical constants that are powers of the base exact when possible.
- State assumptions such as $x > 0$ when a power property introduces $\log x$.
- Do not split logarithms across addition or subtraction inside the argument.

Worked example - annotate the reasoning

Problem. Expand $\log_3 \left(\frac{9x^2}{\sqrt{y}} \right)$ assuming $x > 0$ and $y > 0$.

Model reasoning. $\log_3 9 + 2 \log_3 x - \frac{1}{2} \log_3 y = 2 + 2 \log_3 x - \frac{1}{2} \log_3 y$.

Check your understanding

Independent

Expand $\ln \left(\frac{(x-1)^4 \sqrt{x+2}}{5} \right)$ on its domain.

Concept 2: Condense and preserve the domain

Core idea

Coefficients become exponents; sums become products and differences become quotients.

- Start by moving coefficients into powers.
- Combine only after identifying the individual logarithm restrictions.
- A condensed expression can appear defined at extra inputs; retain the original domain explicitly.

Worked example - annotate the reasoning

Problem. Condense $2 \ln(x-1) - \frac{1}{2} \ln(x+2) + \ln 5$.

Model reasoning. $\ln \left(\frac{5(x-1)^2}{\sqrt{x+2}} \right)$, with original domain $x > 1$.

Check your understanding

Independent

Condense $\log_2 a + 3 \log_2 b - \log_2 c$ for positive a, b, c .

Concept 3: Change of base and graph equivalence

Core idea

$\log_b x = \frac{\ln x}{\ln b}$ allows any valid base to be evaluated or compared.

- Changing the base does not change the function value.
- $\log_b(cx) = \log_b x + \log_b c$ for $c > 0$ and $x > 0$.
- $\log_b(x^p) = p \log_b x$ on the shared domain; this can describe vertical scaling.

Worked example - annotate the reasoning

Problem. Rewrite $\log_5 x$ using natural logarithms and explain why $\log_{25} x = \frac{1}{2} \log_5 x$.

Model reasoning. $\log_5 x = \ln x / \ln 5$. Since $\ln 25 = 2 \ln 5$, $\log_{25} x = \ln x / (2 \ln 5) = \frac{1}{2} \log_5 x$.

Check your understanding

Independent

Compare the graphs of $f(x) = \log_2(8x)$ and $g(x) = \log_2 x$.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

Expand $\log_2(8x^3)$ for $x > 0$.

Exit ticket 2

No calculator

Condense $\ln a - \ln b + 2 \ln c$ for positive variables.

Exit ticket 3

No calculator

Rewrite $\log_7 20$ using natural logarithms.
